HR Analytics – Employee Attrition Analysis

1. Project Overview

This project focuses on analyzing employee attrition using the IBM HR Analytics Employee Attrition & Performance dataset. The objective is to identify employee groups with higher attrition rates and understand patterns across different HR factors.

2. Business Problem

Employee attrition can affect workforce stability and increase recruitment and training costs. Organizations need to understand employee turnover patterns to support better HR planning and employee retention strategies.

This analysis explores employee attrition across departments, age groups, job roles, tenure, overtime, job satisfaction, monthly income, and business travel.

3. Project Objectives

- Calculate key HR metrics, including total employees, employees who left, active employees, and attrition rate.
- Analyze attrition across different employee groups.
- Identify groups with relatively higher attrition rates.
- Build an interactive Excel dashboard to communicate findings.
- Present data-driven observations that can support further HR investigation.

4. Tools and Skills Used

- Microsoft Excel
- Power Query for data cleaning and transformation
- PivotTables for data analysis
- Pivot Charts and regular charts for visualization
- Excel formulas for KPI calculations
- Dashboard design and data storytelling

5. Key Performance Indicators

- Total Employees: 1,470
- Employees Left: 237
- Active Employees: 1,233
- Overall Attrition Rate: 16.12%

6. Analysis Performed

- Department-wise attrition analysis
- Age group-wise attrition analysis
- Tenure-wise attrition analysis
- Overtime-wise attrition analysis
- Job satisfaction-wise attrition analysis
- Monthly income-wise attrition analysis
- Job role-wise attrition analysis
- Business travel-wise attrition analysis

7. Key Findings

- Employees under 25 have the highest age-group attrition rate (39.18%).
- Employees working overtime have a higher attrition rate (30.53%) than employees not working overtime (10.44%).
- Employees with less than two years of tenure have the highest tenure-group attrition rate (34.88%).
- Sales Representatives have the highest job-role attrition rate (39.76%).
- Employees earning below 3,000 monthly have the highest income-group attrition rate (28.61%).

8. Conclusion

The analysis highlights differences in employee attrition across several workforce segments. These findings can help HR teams identify areas that may require further investigation and support data-informed workforce planning and retention strategies.

The observed relationships do not establish that any individual factor causes employee attrition.